

Lock Up Your Data: A Guide to Open Source Encryption

Encryption at rest protects your files when your device is lost or stolen. Encryption in transit protects your data as it moves across the internet. You need both to be truly secure.



Encryption scrambles your digital information so only authorised users can read it. To fully protect your data, you must secure it in two states: at rest (stored on a device) and in transit (moving across the internet). Free, open source tools offer the most reliable way to do this because their code is publicly audited by global security experts to ensure there are no hidden backdoors.

How encryption works

Encryption transforms readable text (plaintext) into scrambled data (ciphertext) using a cryptographic key.

- **Symmetric encryption:** Uses the same key to lock and unlock data. It is fast and ideal for large files. AES-256 is the industry standard.
- **Asymmetric encryption:** Uses a pair of keys—a public key to encrypt and a private key to decrypt. It is slower but allows secure key sharing over the internet.
- **Hybrid encryption:** Modern systems combine both. They use asymmetric encryption to safely share a symmetric key, which then encrypts the actual data. This is how web traffic (TLS) is secured.

Protecting data at rest

Data at rest is information sitting on a hard drive, phone,

USB stick, or cloud server. Without encryption, anyone who physically or digitally accesses the drive can read your files.

The best open source tools for data at rest are:

- **VeraCrypt (Linux, Windows, Mac):** Ideal for encrypting entire hard drives or creating encrypted virtual folders (containers). It supports robust algorithms like AES and Twofish, though the interface has a slight learning curve.
- **LUKS (Linux):** The gold standard for Linux users. It is built directly into the OS, allowing seamless full-disk encryption during system installation.
- **Cryptomator (cross-platform):** Designed specifically for cloud storage (Dropbox, Google Drive, etc). It encrypts your files locally before they upload, ensuring cloud providers cannot read your data.
- **gocryptfs (Linux, Mac, Windows):** A lightweight tool that encrypts individual files and folders rather than entire drives, offering high flexibility.

Protecting data in transit

Data in transit is information travelling across networks. Without protection, attackers on the same network can intercept and read it.

Transport Layer Security (TLS): TLS encrypts connections between browsers and servers, represented by